



Fusing hybrid attentive network with self-supervised dual-channel heterogeneous graph for knowledge tracing

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ABSTRACT

Recently the large-scale influence of Covid-19 promoted the fast development of intelligent tutoring systems (ITS). As a major task of ITS, Knowledge Tracing (KT) aims to capture a student's dynamic knowledge state based on his historical response sequences and provide personalized learning assistance to him. However, most existing KT methods have encountered the data sparsity problem. In real scenarios, an online tutoring system usually has an extensive collection of questions while each student can only interact with a limited number of questions. As a result, the records of some questions could be extremely sparse, which degrades the performance of traditional KT models. To resolve this issue, we propose a Dual-channel Heterogeneous Graph Network (DHGN) to learn informative representations of questions from students' records by capturing both the high-order heterogeneous and local relations. As the supervised learning manner applied in previous methods is incapable of exploiting unobserved relations between questions, we innovatively integrate a self-supervised framework into the KT task and employ contrastive learning via the two channels of DHGN, supplementing as an auxiliary task to improve the KT performance. Moreover, we adopt the attention mechanism, which has achieved impressive performance in natural language processing tasks, to effectively capture students' knowledge state. But the standard attention network is inapplicable to the KT task because the current knowledge state of a student usually shows strong dependency on his recently interactive questions, unlike the situation of language processing tasks, which focus more on the long-term dependency. To avoid the inefficiency of standard attention networks in the KT task, we further devise a novel Hybrid Attentive Network (HAN), which produces both the global attention and the hierarchical local attention to model the long-term and short-term intents, respectively. Then, by the gating network, a student's long-term and short-term intents are combined for efficient prediction. We conduct extensive experiments on several real-world datasets. Experimental results demonstrate that our proposed methods achieve significant performance improvement compared to existing state-of-the-art baselines, which validates the effectiveness of the proposed dual-channel heterogeneous graph framework and hybrid attentive network.

1. Introduction

With the rapid development of adaptive learning, intelligent tutoring systems (ITS) have found more and more applications. ITS may not require the knowledge and experience of education experts as before. By analyzing a student's historical records, ITS can effectively track the dynamic change of his knowledge state. Nowadays knowledge tracing (KT) (Corbett & Anderson, 1994) has become a hot research topic of ITS, which aims to analyze a student's knowledge state and predict his response to a new question. According to the prediction results, teachers can adjust the teaching plans and develop personalized strategies (Adesope et al., 2017) for each student in online education systems.

KT research can be dated back to the late 1960s. At the early stage, some novel Bayesian modeling methods (Pozna et al., 2010; Yudelson et al., 2013) were developed to make great progress in the heuristic modeling. Bayesian Knowledge Tracing (BKT) (Kasurinen & Nikula, 2009) is a typical class of methods. BKT represents a student's knowledge acquisition level with a binary number (1 for mastery and 0 for not mastery). Later, multiple factors are taken to construct a student's knowledge state. The Difficulty, Ability, and Student History (DASH) model (Lindsey et al., 2014) considers the impact of additional factors, such as question difficulty, error rate, response time, on students' final response. However, the above models strongly depend on the statistical estimation of parameters and are not robust enough.

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Over the past decade, there has emerged some attempts to construct nonlinear modeling approaches (Fersini et al., 2014; Hedrea & Petriu, 2021), which achieved superior performance in diverse fields. Along this trend, neural network-based approaches were applied to knowledge tracing. Due to their strong robustness and portability, neural network-based methods generally perform better than traditional ones. For example, Deep Knowledge Tracing (DKT) (Piech et al., 2015) first utilizes the long short term memory (LSTM) network (Hochreiter & Schmidhuber, 1997) to track a student's state evolution process. Ensemble Knowledge Tracing (EKT) (Sun et al., 2022) introduces an ensemble learning into KT, which fuses BKT and DKT, and extends AdaBoost to deal with KT sequential data. But this recurrent network architecture is insufficient to capture long-term dependencies and is easy to lose early semantic details. Then some memory-based methods (Abdelrahman & Wang, 2019; Liu et al., 2021; Zhang et al., 2017), like Dynamic Key-Value Memory Networks for Knowledge Tracing (DKVMN) (Zhang et al., 2017), were proposed to resolve the above issues by introducing an external memory matrix.

Later, attention-based methods (Vaswani et al., 2017) have made great advances in handling sequential data, which can automatically study different weights for each interaction to capture the related information in the historical sequences. The Self-Attentive Knowledge Tracing (SAKT) (Pandey & Karypis, 2019) first utilizes the attention mechanism to focus on the historical interactions that are highly relevant to the target question. However, due to the long recording cycle of KT, the traditional attention mechanism does not perform well. Context-Aware Attentive Knowledge Tracing (AKT) (Ghosh et al., 2020) introduces a monotonic attention mechanism that integrates the exponential decay and characterizes the importance of memory decay in KT.

Recently, graph neural network (GNN) (Dong et al., 2023, 2022; Kipf & Welling, 2016; Schlichtkrull et al., 2018; Wang et al., 2019) shows superior performance on the relation network and graph-structured data, and can encode rich context information by aggregating graph attributive and structural features. Some GNN-based methods (Ni et al., 2023; Song et al., 2021; Wang et al., 2022; Wu et al., 2022; Yang et al., 2020) have been leveraged in the KT field, like A Graph-based Interaction Model for Knowledge Tracing (GIKT) (Yang et al., 2020), which combines the latent relations between questions and skills by utilizing the graph convolutional network (GCN) (Kipf & Welling, 2016). A joint graph convolutional network based Deep Knowledge Tracing (JKT) (Song et al., 2021) models the multi-dimensional relationships of "exercise-to-exercise" and "concept-to-concept" via two graph networks.

Although existing methods have achieved significant improvement in knowledge tracing, they still suffer from the following limitations.

1. The performance of those methods mainly relies on the scale of historical interactions. However, due to the sparsity of records in KT, the visiting frequency of some questions is extremely low. To avoid over-fitting and over-parameterization, previous methods (Ghosh et al., 2020; Piech et al., 2015; Zhang et al., 2017) usually select the skill records, instead of the question records (the number of questions is generally much larger than that of skills), to model a student's knowledge state. But this usually leads to the loss of semantic details being contained in historical interactions.
2. Some graph-based methods (Song et al., 2021; Tong et al., 2020; Yang et al., 2020) do improve the representations of questions by fusing the graph-structured information, but may mostly overlook the heterogeneity of nodes. In KT, these recorded questions usually attach diverse attributes, like question type, difficulty coefficient, involved skill, etc. Obviously, simple homogeneous graphs (question-skill pair) are incapable of characterizing heterogeneous relations and building expressive representations for questions.

3. Traditional attention-based methods often yield poor performance in long sequence prediction (Pandey & Karypis, 2019). In fact, memory decays over time, i.e., the knowledge, which was well mastered before, may be forgotten now (Carpenter et al., 2008; Nagatani et al., 2019). It is non-trivial to balance the effects of short-term and long-term memory in student modeling.

To address the above limitations, we propose a novel knowledge tracing model, a Hybrid Attentive Network fused with Dual-channel Heterogeneous Graph Network (HAN-DHGN). First, we introduce a dual-channel heterogeneous graph network (DHGN) and then learn question embeddings via the network schema and meta-path channels of DHGN, which can effectively exploit the local and high-order relations between questions. As most KT methods are based on a supervised learning paradigm, which is limited by the data sparsity in real scenarios, making it incapable to learn quality representations, we introduce self-supervised learning into DHGN, aiming to exploit unobserved relations and reinforce the question representations. Specifically, we utilize the question representations learned by the two channels of DHGN to conduct contrastive learning, which maximizes the similarity between positive pairs while minimizing the similarity between negative-sampled counterparts. To the best of our knowledge, the potential of contrastive learning for heterogeneous graph representation learning and KT has not yet been explored. Therefore, our work aims to fill this research gap and investigate the effectiveness of contrastive learning in the KT field.

Moreover, to improve the efficiency of traditional attention-based models and perceive the dynamics of memory, we propose a novel hybrid attentive network (HAN), including both global attention and hierarchical local attention, to track a student's intents in the long and short terms, respectively. In HAN, the global attention deals with the whole interactive sequences to capture the long-term intent. To characterize the memory decay in the long sequence, we add an exponential decay to the standard attention network. The local attention focuses on the recent interactive details by a cluster of hierarchical attention windows. As the window size increases with the number of layers, our model can perceive the relevance in longer scope and extract multi-scale semantics features. The hierarchical construction greatly optimizes the short-term intent modeling. Then the two intents are adaptively integrated via a gating network for the prediction task. Finally, we unify the prediction task and the self-supervised task, and leverage the multi-task training strategy to jointly optimize the two tasks. Our contributions can be summarized as follows.

- We propose a novel dual-channel heterogeneous graph network (DHGN) for representation learning. The dual-channel framework can effectively exploit the local and high-order relations between questions, helping the model build expressive question representations for the downstream prediction task. Although the existing GNN-based methods demonstrate their superiority in the KT field, most of them are built upon a simple homogeneous graph, which neglects the complex heterogeneity between questions and may consequently fail to learn expressive representations for questions.
- We devise a novel hybrid attentive network (HAN) to capture a student's long-term and short-term intents simultaneously. The conventional attention mechanism, which focuses on long-term dependencies, is incapable of modeling memory decay in real scenarios and often yields poor performance. Compared with the conventional attention mechanism, our HAN fuses global attention and hierarchical local attention, aiming to model a student's state in a more effective manner, and achieves decent gains.
- We innovatively integrate a self-supervised task into the training of our DHGN to enhance heterogeneous graph modeling and further improve the KT prediction performance. Existing KT methods are usually built on a supervised learning paradigm and their performances are constrained by the data sparsity problem. So it is difficult for these methods to learn high-quality representations. Conversely, the

self-supervised learning adopted in this paper can find supervised signals from the input data itself and exploit unobserved relations spontaneously to reinforce the question representation.

- We conduct extensive experiments on several real-world datasets and the results demonstrate the performance superiority of our HAN-DHGN against some state-of-the-art KT methods.

The rest of this paper is organized as follows. Section 2 briefly reviews the related work. Section 3 formulates the concerned KT task and presents the proposed HAN-DHGN in detail. We conduct extensive experiments on four real-world datasets and comprehensively compare HAN-DHGN with some state-of-the-art methods in Section 4. Finally, Section 5 concludes the paper and comments on the limitation of the proposed HAN-DHGN and the future work.

2. Related work

2.1. Knowledge tracing

In recent years, the development of neural network (NN) promotes considerable advances of knowledge tracing (KT). KT aims to capture a student's knowledge state according to his interactive questions and predict his response to a new question. DKT (Piech et al., 2015), the first attempt to employ RNN-based methods in KT, utilizes the LSTM network to model the interactive question sequences. DKVMN (Zhang et al., 2017), DSCMN (Minn et al., 2019) and SKVMN (Abdelrahman & Wang, 2019) leverage the memory-based network to capture long-term dependencies of historical records. HMN (Liu et al., 2021) devises a hierarchical memory network to well fit human memory mechanism in KT. Additionally, the attention-based methods (Ghosh et al., 2020; Jiang et al., 2023; Pandey & Karypis, 2019; Wei et al., 2022), like SAKT (Pandey & Karypis, 2019), AKT (Ghosh et al., 2020), introduce attention mechanisms to emphasize the historical interactions that are highly relevant to the target question. TISAKT (Wei et al., 2022) integrates the time interval between two consecutive interactions into self-attention mechanisms. HawkesKT (Wang, Ma et al., 2021) investigates fine-grained temporal cross-effects between different skills in KT and explicitly models these cross-effects by the point process. To enrich question representations, additional attributes are considered. EERNNA (Su et al., 2018) encodes question embeddings with the content of questions. But it is impractical to collect the content information in real scenarios. GIKT (Yang et al., 2020) builds a relation graph to extract the underlying relations between questions and skills. Along this line, CQKT (Wang et al., 2022) proposes a calibrated Q-matrix-based framework to track knowledge proficiency of students dynamically. HHN (Wu & Ling, 2023) develops a hypergraph network to capture the complex beyond-pairwise semantic relations among questions. SGKT (Wu et al., 2022) constructs an exercise session graph to model a student's answering process and takes the self-attention with a forgetting mechanism to model exercises and skills. Although these recurrent and attentive methods have demonstrated their superiority in the KT field, they are limited by the data sparsity problem, making it difficult to learn high-quality representations for students' historical interactions. Although Graph Neural Network (GNN)-based methods can alleviate the aforementioned issue to some extent, they fail to capture the complex and higher-order correlations between questions. To this end, we propose a novel graph structure to model the heterogeneity omitted by previous methods.

2.2. Heterogeneous graph neural network

Recently, the heterogeneous graph has caught much research attention, and shown its great capability of modeling various types of nodes and relations. The meta-paths of Heterogeneous Information Network (HIN) are usually described as semantic relations between nodes pairs. By aggregating features from different meta-paths, the heterogeneous structure can learn more comprehensive presentations for nodes than

a homogeneous graph. For example, HAN (Wang et al., 2019) utilizes hierarchical attention to capture the node-level and semantic-level semantics. Based on this fact, DisenHAN (Wang et al., 2020) proposes a disentangled heterogeneous graph attention network for recommendation, which considers the entanglement of latent factors stemming from different meta-paths. HGT (Hu et al., 2020) proposes a heterogeneous graph transformer architecture for modeling Web-scale heterogeneous graphs. HetGNN (Zhang et al., 2019) uses LSTM networks to aggregate feature information of sampled neighboring nodes to obtain the ultimate node embedding. MetaHIN (Lu et al., 2020) develops a meta-learning paradigm based on HIN to resolve the cold-start problem of recommendation systems. Generally, the performance of GNNs is vulnerable to noisy information. To resolve this issue, DNAG (Dong et al., 2023) presents a novel de-noising aggregator, which utilizes Principal Component Analysis (PCA) to preserve the aggregated real signals and simultaneously filter out the Gaussian noise in GNNs. Dong et al. (2022) innovatively introduces an injective aggregation function to improve the representation capability of GNNs. Unfortunately, there is still a gap between heterogeneous graph neural network and KT. We want to fill this gap by constructing the heterogeneous graph between questions and then propose a dual-channel heterogeneous graph network to aggregate neighbors' information and enhance the question representation.

2.3. Contrastive learning

As an emerging paradigm in self-supervised learning, contrastive learning aims to learn discriminative representations by contrasting positive and negative pairs. It was firstly used in visual representation learning, in which negative samples are generated by cropping, resizing, rotation, etc. Later, the contrastive framework is extended to enhance graph representation learning via self-discrimination. DGI (Velickovic et al., 2019) is the first to introduce the InfoMax principle into GNN and augment the original graph by simply shuffling node features. InfoGraph (Sun et al., 2019) extends deep InfoMax to learn graph-level representations. GMI (Peng et al., 2020) measures MI between a center node and its local patch from node-level and topology-level features. MVGRL (Hassani & Khasahmadi, 2020) applies a graph diffusion network (GDN) to generate additional structural views. GCA (Zhu et al., 2021) optimizes the uniform data augmentation schemes with the adaptive mechanism. Although contrastive learning has been widely implemented in diverse fields, its potential for the KT task has not been explored. So we plan to integrate contrastive learning into the scenarios of KT and heterogeneous graph modeling.

3. Methodology

In this section, we formulate the concerned KT problem and introduce the proposed model in detail.

3.1. Problem statement

In the KT task, each student's historical records consist of a sequence of questions and responses. Denote the set of questions as Q . Define the sequence of records of a student as $S_t = \{(q_1, r_1), (q_2, r_2), \dots, (q_t, r_t)\}$ in the chronological order. $q_t \in Q$ records the t th question interacted by the student and $r_t \in \{0, 1\}$ is the corresponding response (1 stands for correct and 0 for incorrect).

Definition 3.1.1 (Heterogeneous Graph Shi et al., 2016; Sun & Han, 2013). A heterogeneous graph is defined as $G = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the node set and $\mathcal{E}$ is the edge set. There are two mapping functions in the heterogeneous graph, including node type mapping function $\phi : \mathcal{V} \rightarrow \mathcal{A}$ and edge type mapping function $\psi : \mathcal{E} \rightarrow \mathcal{R}$, where $\mathcal{A}$ and $\mathcal{R}$ represent the node type set and the edge type set, respectively, and satisfy $|\mathcal{A}| + |\mathcal{R}| > 2$.

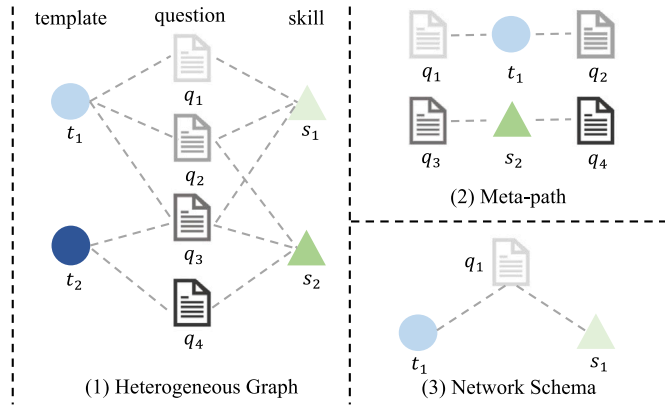


Fig. 1. An illustration of the heterogeneous graph for KT, which includes three types of nodes and two types of meta-paths. For simplicity, we only show two additional attributes, including skill and template.

Definition 3.1.2 (Meta-path Shang et al., 2016; Sun et al., 2011). In the heterogeneous graph, the meta-path Φ is usually defined in the form of $\mathcal{A}_1 \xrightarrow{R_1} \mathcal{A}_2 \xrightarrow{R_2} \dots \xrightarrow{R_{l-1}} \mathcal{A}_l$. The two node type sets $\mathcal{A}_1$ and $\mathcal{A}_l$ are connected by the composite relation $R = R_1 \circ R_2 \dots R_{l-1}$, where $\circ$ represents the composition operator on relations.

Definition 3.1.3 (Network Schema Wang, Liu et al., 2021). The network schema is usually defined as $T_G = (\mathcal{A}, \mathcal{R})$, which depicts a directed graph defined over the node type set $\mathcal{A}$, with edges being relations from $\mathcal{R}$.

Based on the above definitions, we embody the heterogeneous graph upon the KT task. As shown in Fig. 1(1), the heterogeneous graph consists of different node types (template (T), question (Q), skill (S)) and relation types ($Q \rightarrow S$ means the question involves this skill and $Q \rightarrow T$ means the question belongs to this template). From Fig. 1(2), we can distinctly find that two questions are connected by different meta-paths, where Q-T-Q, Q-S-Q represent that two different questions cover the same template and skill. The multiple meta-paths in HIN indicate multifaceted semantics, which represent high-order heterogeneous relations between nodes. In Fig. 1(3), the network schema shows question q_1 involves skill s_1 and belongs to template t_1 simultaneously. The direct connections between different types of nodes depicted in the network schema are usually viewed as the local relations.

Based on the above notation, the task of knowledge tracing can be formulated as follows.

Given a student's interactive sequence $S_t = \{(q_1, r_1), (q_2, r_2), \dots, (q_t, r_t)\}$ and the pre-defined heterogeneous graph between questions G , our task is to learn $f(\cdot)$ to predict his next response r_{t+1} to a new question q_{t+1} , which is formulated as

$$[S_t, G, q_{t+1}] \xrightarrow{f} r_{t+1}. \quad (1)$$

3.2. Dual-channel heterogeneous graph network

This section introduces the proposed Dual-channel Heterogeneous Graph Network (DHGN), which aims to capture both the high-order and local relations via the meta-path and network schema channels, and then learn an informative representation for the question (see Fig. 2).

3.2.1. Meta-path channel

Here we encode the high-order relations in the meta-path channel. Generally, multiple meta-paths of HIN reflect the semantics details from diverse aspects. To explicitly perceive the discrepancy of semantics, we implement the information aggregation independently for different meta-paths. Suppose there are K types of meta-paths starting from

target node i , denoted as $\{\Phi_1, \Phi_2, \dots, \Phi_K\}$, and its neighbors based on meta-path Φ_k are tagged as $\mathcal{N}_i^{\Phi_k}$. For example, as shown in Fig. 1(1), for meta-path Q-T-Q, the question node q_1 consists of two neighbors, q_2 and q_3 .

The initial node embeddings are denoted as $\{\mathbf{h}_1^0, \mathbf{h}_2^0, \dots, \mathbf{h}_{|\mathcal{V}|}^0\} \in \mathbb{R}^{|\mathcal{V}| \times D}$, where D is the dimension of embeddings. Each of them is a trainable embedding associated with the question ID. Then we utilize the attention mechanism to capture the relevance within the single meta-path Φ_k . The importance of neighbor node $j \in \mathcal{N}_i^{\Phi_k}$ to target node i (Veličković et al., 2017) is calculated as

$$\eta_{ij}^{\Phi_k} = \text{Att}(\mathbf{h}_i^{l-1}, \mathbf{h}_j^{l-1}, \Phi_k) = \psi(\mathbf{a}_{\Phi_k}^T [\mathbf{W}_i^{\Phi_k} \mathbf{h}_i^{l-1} \parallel \mathbf{W}_j^{\Phi_k} \mathbf{h}_j^{l-1}]), \quad (2)$$

where $\mathbf{a}_{\Phi_k}^T, \mathbf{W}_i^{\Phi_k}, \mathbf{W}_j^{\Phi_k}$ represent the path-specific weight vector and projection matrices for meta-path Φ_k . These parameters are shared at each layer to avoid over-fitting. $\mathbf{h}_i^{l-1}$ denotes the refined embedding of node i after $l-1$ layers propagation. ψ is the LeakyReLU activation function and $\parallel$ stands for the concatenation operation.

Usually, those attention weights are normalized via softmax function, which returns positive values and dense output probabilities. However, this nonzero probabilities may assign weights to some useless features and lead to the incapability of finding relevant nodes. To address this limitation, we introduce the following novel transformation function, α -entmax (Peters et al., 2019), to replace the softmax,

$$\alpha\text{-entmax} = \arg\max_{p \in \Delta^d} p^T \mathbf{x} + H_\alpha^T(p), \quad (3)$$

$$H_\alpha^T(p) = \begin{cases} \frac{1}{\alpha(\alpha-1)} \sum_j (p_j - p_j^\alpha), & \alpha \neq 1 \\ H^S(p), & \alpha = 1 \end{cases} \quad (4)$$

where $H_\alpha^T(p)$ is Tsallis α -entropies (Tsallis, 1988), which is a family of entropies parametrized by a scalar $\alpha > 1$. Tsallis entropies interpolate between the Shannon and Gini entropies. Based on the above equation, we can find that 1-entmax $\equiv$ softmax and 2-entmax $\equiv$ sparsemax. For $1 < \alpha < 2$, entmax yields a function family continuously interpolating between softmax and sparsemax, which inherits the desirable sparsity of sparsemax, and shows smoother corners.

In this paper, we adopt the α -entmax in the attention mechanism to eliminate the impact of unrelated neighbor nodes. Then the weight coefficient is calculated as

$$\beta_{ij}^{\Phi_k} = \alpha\text{-entmax}(\eta_{ij}^{\Phi_k}). \quad (5)$$

To reflect the discrepancy of meta-paths in the feature transformation, we propose a method to learn α adaptively based on those connected nodes in the current meta-path. Specifically, we utilize a pooling function, which averages all node embeddings within meta-path Φ_k , to generate its α as

$$\alpha = \sigma \left\{ \frac{1}{|\mathcal{N}_i^{\Phi_k}|} \sum_{j \in \mathcal{N}_i^{\Phi_k}} (\mathbf{W}_{\Phi_k} \cdot \mathbf{h}_{j, \Phi_k}^{l-1} + \mathbf{b}_{\Phi_k}) \right\} + 1, \quad (6)$$

where $\mathcal{N}_i^{\Phi_k}$ represents the set of nodes in meta-path Φ_k . $\mathbf{W}_{\Phi_k}$ and $\mathbf{b}_{\Phi_k}$ are the path-specific weight matrix and the bias vector, respectively. σ denotes the sigmoid activation function and the value of α is mapped into the interval of $[1, 2]$. Note that $\mathbf{h}_{j, \Phi_k}^0 = \mathbf{h}_j^0$ when $l = 1$ (see Fig. 3).

Next, we utilize the above weight coefficient to fuse the features of neighbor nodes in meta-path Φ_k as

$$\mathbf{h}_{i, \Phi_k}^l = \sum_{j \in \mathcal{N}_i^{\Phi_k}} (\beta_{ij}^{\Phi_k} \cdot \mathbf{h}_j^{l-1}). \quad (7)$$

After separately aggregating the information from neighbors based on different meta-paths, we can obtain the heterogeneous semantics representations $\{\mathbf{h}_{i, \Phi_1}^l, \mathbf{h}_{i, \Phi_2}^l, \dots, \mathbf{h}_{i, \Phi_K}^l\}$ for target node i . Furthermore, we combine them to form the node embedding at l th layer as

$$\mathbf{h}_i^l = \mathbf{W}_K \cdot \left[\parallel_{k=1}^K \mathbf{h}_{i, \Phi_k}^l \right], \quad (8)$$

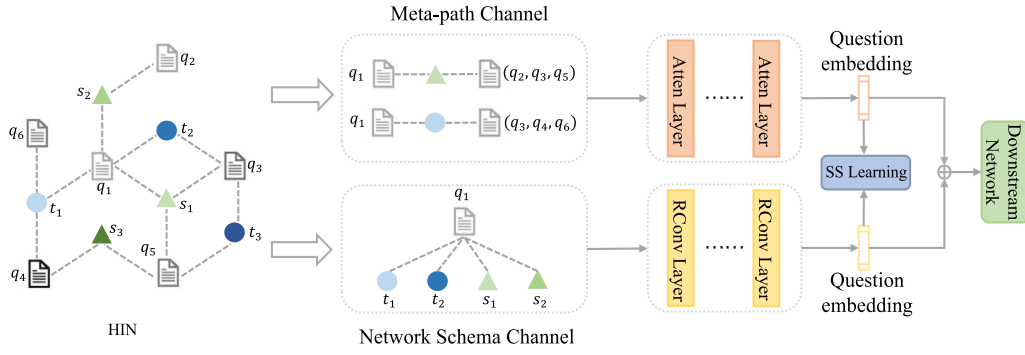


Fig. 2. The architecture of DHGN. The question embedding is learned through the meta-path and network schema channels. Then the two representations are utilized to perform self-supervised learning and are fed into the downstream network.

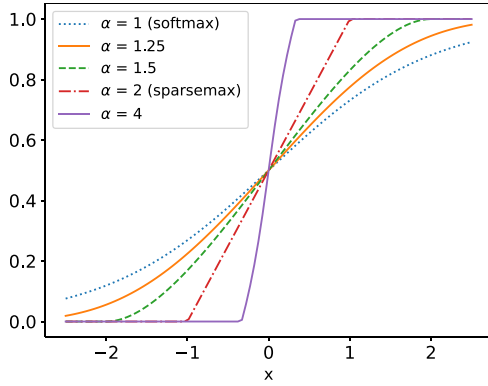


Fig. 3. Illustration of entmax in the two-dimensional case.

where $\mathbf{W}_K$ is the weight matrix. Last, we average the node embeddings obtained at each layer to get the final representation of node i in the meta-path channel as

$$\mathbf{h}_i^\Phi = \frac{1}{L_\Phi + 1} \sum_{l=0}^{L_\Phi} \mathbf{h}_i^l. \quad (9)$$

3.2.2. Network schema channel

Here we aim to extract the intrinsic dependencies from the network schema to learn the local representation of nodes. Due to the heterogeneity of nodes, the target node usually contains various neighbors. Obviously, different types of neighbors are of different importance to the target node. Just equally treating them may not be reasonable. So we introduce the relational GCN (RGCN) (Schlichtkrull et al., 2018) to capture the local relations in the network schema channel. Similar to the above, we assume that there are M types of neighbors for target node i , as $\{\Psi_1, \Psi_2, \dots, \Psi_M\}$ and its neighbors can be tagged as $\mathcal{N}_i^{\Psi_m}$ with type Ψ_m . An example is illustrated in Fig. 1(2), where question node q_1 contains two types of neighbors, t_1 and s_1 , based on the network schema. The RGCN is then defined as

$$\mathbf{h}_i^l = \psi \left\{ \mathbf{W}_i \mathbf{h}_i^{l-1} + \frac{1}{M} \cdot \frac{1}{|\mathcal{N}_i^{\Psi_m}|} \sum_{m \in M} \sum_{j \in \mathcal{N}_i^{\Psi_m}} (\mathbf{W}_{\Psi_m} \cdot \mathbf{h}_j^{l-1}) \right\}, \quad (10)$$

where $\mathbf{W}_i, \mathbf{W}_{\Psi_m}$ are the weight matrices and ψ is the LeakyReLU activation function. Likewise, we obtain the representation of node i in the network schema channel as

$$\mathbf{h}_i^\Psi = \frac{1}{L_\Psi + 1} \sum_{l=0}^{L_\Psi} \mathbf{h}_i^l. \quad (11)$$

Last, we sum the question node embeddings obtained in the meta-path and network schema channels to get the final representation of

question i as

$$\mathbf{h}_i = \mathbf{h}_i^\Psi + \mathbf{h}_i^\Phi. \quad (12)$$

3.3. Contrastive learning

The two presentations to be obtained in the meta-path and network schema channels encode the high-order and local structural information, respectively. They know little about each other, but can mutually complement. Following Ren et al. (2019) and Velickovic et al. (2019), the two channels of DHGN are treated as two views, which can be the ground-truth of each other for contrastive learning. We treat the views of the same node as the positive pairs and the views of any different nodes as the negative pairs naturally. Formally, we utilize InfoNCE objective (Van den Oord et al., 2018) as the contrastive loss to contrast node representations from different views. The objective of the meta-path channel of node i is defined as

$$\mathcal{L}_i^\Phi = -\log \frac{\exp(s(\mathbf{h}_i^\Phi, \mathbf{h}_i^\Psi)/r)}{\exp(s(\mathbf{h}_i^\Phi, \mathbf{h}_i^\Psi)/r) + \sum_{j \neq i} \exp(s(\mathbf{h}_i^\Phi, \mathbf{h}_j^\Phi)/r) + \sum_{j \neq i} \exp(s(\mathbf{h}_i^\Phi, \mathbf{h}_j^\Psi)/r)}, \quad (13)$$

where $s(\cdot)$ is the cosine similarity function to measure the similarity between two vectors and r denotes a temperature parameter, and $\mathbf{h}_i^\Phi, \mathbf{h}_i^\Psi$ represent the embeddings of question i obtained in the meta-path and network schema channels, respectively.

Considering the symmetry of the two views, we can define the contrastive loss of network schema channel $\mathcal{L}_i^\Psi$ analogously. Combining these two losses yields the overall objective of the self-supervised task,

$$\mathcal{L}_{ssl} = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} (\mathcal{L}_i^\Phi + \mathcal{L}_i^\Psi). \quad (14)$$

3.4. State extractor network

Here we utilize the RNN with Gated Recurrent Units (GRU) (Chung et al., 2014), which has shown its remarkable capability in capturing sequential intents, to trace a student's knowledge state according to his historical sequence. The dependencies between interactive questions and the dynamics of students' response are tracked by a parallel GRU independently as

$$\mathbf{x}_{q_t} = \text{GRU}_q(\mathbf{h}_{q_1}, \mathbf{h}_{q_2}, \dots, \mathbf{h}_{q_t}), \mathbf{x}_{q_t} \in \mathbb{R}^{D_r}, \quad (15)$$

$$\mathbf{x}_{r_t} = \text{GRU}_r(\mathbf{h}_{r_1}, \mathbf{h}_{r_2}, \dots, \mathbf{h}_{r_t}), \mathbf{x}_{r_t} \in \mathbb{R}^{D_r}, \quad (16)$$

where D_r denotes the size of the hidden layer of GRU, $\mathbf{h}_{q_t}$ is the obtained embedding of question q_t that the student interacts with at the t th step, and $\mathbf{h}_{r_t}$ is the corresponding response embedding, indicating

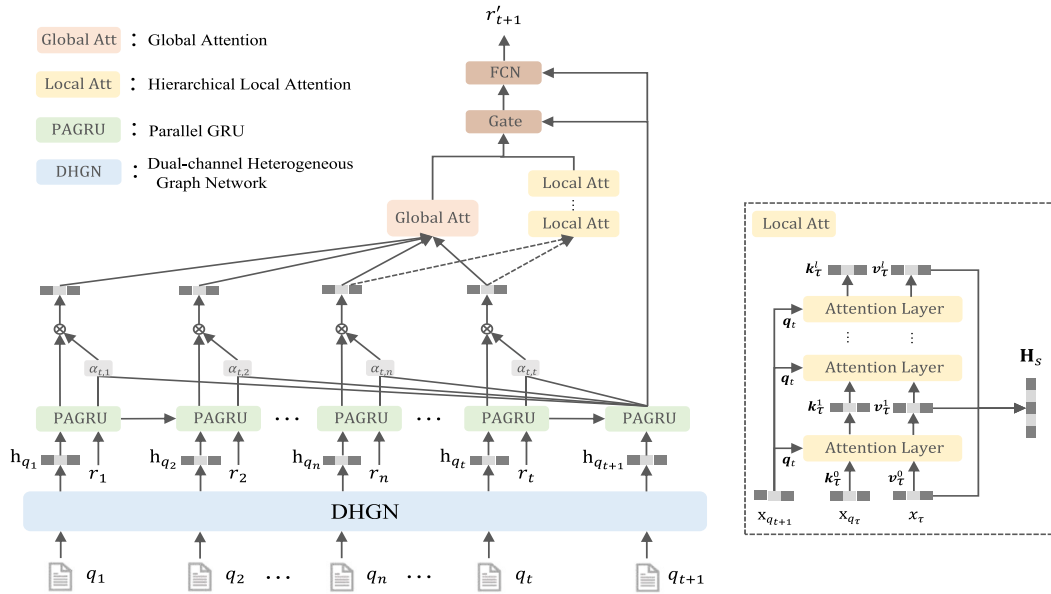


Fig. 4. An overall view of HAN-DHGN and the construction of the hierarchical local attention network. Pa-GRU means the proposed parallel GRU of the state extractor network.

correct or incorrect. The knowledge state $\mathbf{x}_t$ at the t th step is produced by the summation of two hidden states as

$$\mathbf{x}_t = \mathbf{x}_{q_t} + \mathbf{x}_{r_t}. \quad (17)$$

The parallel mode alleviates the semantic blending caused by the coupling between the question and response embeddings under previous methods (Piech et al., 2015), and can better extract a student's unbiased state for prediction to some extent (see Fig. 4).

3.5. Hybrid attention network

To improve the inefficiency of traditional attention networks over long sequences, we propose a novel Hybrid Attention Network (HAN), including global attention for long-term intent modeling and hierarchical local attention for short-term intent modeling, which can effectively capture the dynamics over time. Finally, the two intents are adaptively integrated via a neural gating network for prediction.

3.5.1. Global attention network

To focus on the interactions being strongly related to the target question and retain the long-range dependencies that may be lost in the recurrent network iterations, we apply a standard attention network on the whole sequence and learn the long-term intent representation. Suppose the query is given as $\mathbf{q}_t = \mathbf{x}_{q_{t+1}}$, the key as $\mathbf{k}_\tau = \mathbf{x}_{q_\tau}$ and the value as $\mathbf{v}_\tau = \mathbf{x}_\tau$, $\tau \leq t$.

The importance of the interaction at the τ th step to target question $\mathbf{q}_{t+1}$ is calculated via the softmax function as

$$\alpha_{t,\tau} = \text{Softmax}\left(\frac{\mathbf{q}_t \mathbf{k}_\tau^T}{\sqrt{D_r}}\right) = \frac{\exp\left(\frac{\mathbf{q}_t \mathbf{k}_\tau^T}{\sqrt{D_r}}\right)}{\sum_{\tau'=1}^t \exp\left(\frac{\mathbf{q}_t \mathbf{k}_{\tau'}^T}{\sqrt{D_r}}\right)}, \quad (18)$$

where the scale factor $\sqrt{D_r}$ is used to normalize the inner product. Considering the long learning cycle of KT, the knowledge, which was well mastered before, may be forgotten now. The memory decay has a significant impact on students' knowledge state. To emphasize the importance of memory decay in modeling long interaction sequences, we introduce an exponential decay, which takes the time interval into

account when calculating the correlation between two interactions. The decay coefficient $\delta_{t,\tau}$ is computed as

$$\delta_{t,\tau} = \exp(-\beta|t - \tau|), \quad (19)$$

where β is the decay rate. As mentioned above, the input of the attention network comes from the output of the extractor network. Although the parallel GRU can effectively extract the dependencies, this recursive structure tends to cause the loss of semantic details. Additionally, to retain the details, we add the following semantic-aware coefficient $\eta_{t,\tau}$, which encodes the low-layer embedding features (Ji et al., 2020) as

$$\eta_{t,\tau} = \sigma(\mathbf{h}_{q_{t+1}} \mathbf{W}_\eta \mathbf{h}_{q_\tau}^T + \mathbf{b}_\eta), \quad (20)$$

where $\mathbf{W}_\eta$ is the weight matrix and $\mathbf{b}_\eta$ is the biased vector, σ represents the sigmoid activation function. The refined coefficient $\zeta_{t,\tau}$ is calculated as

$$\zeta_{t,\tau} = \delta_{t,\tau} \cdot \eta_{t,\tau} \cdot \alpha_{t,\tau}. \quad (21)$$

Finally, we generate the global intent representation $\mathbf{H}_l$ based on the entire historical sequences as

$$\mathbf{H}_l = \sum_{\tau=1}^t \zeta_{t,\tau} \mathbf{v}_\tau. \quad (22)$$

3.5.2. Hierarchical local attention network

To encode the short-term intent, an intuitive way is to only consider a couple of recent interactions within a fixed window. However, it is difficult to balance the impact of interactive details and the scope. To address this limitation, we propose a hierarchical local attention network, and generate the multi-scale short-term representation. Similarly we calculate the attention scores at the l th layer as

$$\alpha_{t,\tau}^l = \text{Softmax}\left(\frac{\mathbf{q}_t \mathbf{k}_\tau^{l-1,T}}{\sqrt{D_r}}\right), \quad \tau \in [t - w_0 \cdot \gamma^{l-1}, t], \quad (23)$$

where w_0 is the initial window size at the first layer and γ is the dilated coefficient. Note that the attention window size increases layer by layer. At lower layers, smaller windows produce more interactive details while at higher layers, larger windows enable wider receptive fields for interactions and help model perceive the relevance in the

longer scope. Given $q_t = \mathbf{x}_{q_{t+1}}$, $\mathbf{k}_t^0 = \mathbf{x}_{q_t}$ and $\mathbf{v}_t^0 = \mathbf{x}_t$, key and value are updated as

$$\mathbf{k}_t^l = \sum_{\tau=t-w_0, \gamma^{l-1}}^t \alpha_{t,\tau}^l \mathbf{k}_\tau^{l-1}, \quad (24)$$

$$\mathbf{v}_t^l = \sum_{\tau=t-w_0, \gamma^{l-1}}^t \alpha_{t,\tau}^l \mathbf{v}_\tau^{l-1}. \quad (25)$$

Finally, we generate the local intent representation $\mathbf{H}_s$ by combining the multi-scale semantics in different layers as

$$\mathbf{H}_s = \mathbf{W}_s \cdot [\| \mathbf{v}_t^l \|]_{l=0}^{L_a}, \quad (26)$$

where $\mathbf{W}_s$ is the weight matrix and L_a is the number of layers in the local attention.

3.5.3. Neural gating network

As the long-term and short-term intents may contribute differently to the response prediction corresponding to target question q_{t+1} , we utilize a neural gating network to combine them adaptively as

$$G_t = \sigma(\mathbf{W}_l[\mathbf{H}_l \| \mathbf{h}_{q_{t+1}}] + \mathbf{W}_s[\mathbf{H}_s \| \mathbf{h}_{q_{t+1}}]), \quad (27)$$

$$\mathbf{V}_t = G_t \cdot \mathbf{H}_l + (1 - G_t) \cdot \mathbf{H}_s, \quad (28)$$

where G_t is the gate vector obtained by the long-term and short-term intent embeddings, σ indicates the sigmoid activation function, and $\mathbf{W}_l, \mathbf{W}_s$ is the weight matrix. The final state embedding $\mathbf{V}_t$ can well balance the long-term and short-term incentives

3.6. Prediction network

To leverage the low-layer semantics explicitly, we concatenate the final state and the target question embedding (Ghosh et al., 2020). Next, we utilize a fully-connected network (FCN) to generate the predicted probability $r'_{t+1} \in [0, 1]$ according to which the student can correctly answer the target question. For training these learnable parameters, we adopt the following binary cross-entropy loss as the objective function of prediction task,

$$r'_{t+1} = \sigma(\text{FCN}(\mathbf{V}_t \| \mathbf{h}_{q_{t+1}})), \quad (29)$$

$$\mathcal{L}_{main} = - \sum (r_{t+1} \log(r'_{t+1}) + (1 - r_{t+1}) \log(1 - r'_{t+1})), \quad (30)$$

where $r'_{t+1} \in [0, 1]$ and $r_{t+1} \in \{0, 1\}$ denote the predicted and true response at step $t+1$, respectively, σ is the sigmoid activation function.

Finally, we unify the main prediction task and the self-supervised task, and optimize the two tasks with a multi-task training strategy (Collobert & Weston, 2008). Formally, the joint objective is defined as

$$\mathcal{L} = \mathcal{L}_{main} + \lambda \mathcal{L}_{ssl}, \quad (31)$$

where λ is the hyperparameter to control the contribution of the self-supervised task. The overall process and modeling steps of our method are presented in Algorithm 1.

4. Results and validation

In this section, we take several public datasets to evaluate the performance of our proposed model and some state-of-the-art models. Particularly, two versions of the proposed DHGN are investigated in the experiments, where DHGN represents the vanilla version and S^2 -DHGN represents the self-supervised version. Furthermore, we conduct some ablation studies to verify the effectiveness and superiority of different components of our model.

Algorithm 1: The modeling steps of our method

Input : The heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$,
The meta-path and $\{\Phi_1, \Phi_2, \dots, \Phi_K\}$,
The layer of meta-path channel L_Φ ,
The layer of network schema channel L_Ψ ,
The observed sequence $\mathcal{S}_t = \{(q_1, r_1), (q_2, r_2), \dots, (q_t, r_t)\}$,
The target question q_{t+1} .

Output: The predicted response r'_{t+1} corresponding to q_{t+1} .

```

1 for  $i \in \mathcal{V}$  do
2   for  $l \in L_\Phi$  do
3     for  $\Phi_k \in \{\Phi_1, \Phi_2, \dots, \Phi_K\}$  do
4       Find the meta-path based neighbors  $\mathcal{N}_i^{\Phi_k}$ ;
5       for  $j \in \mathcal{N}_i^{\Phi_k}$  do
6         Calculate the coefficient  $\eta_{ij}^{\beta_k}$  via Equ.(5);
7         Generate the node representation  $\mathbf{h}_{i,\Phi_k}^l$  via
           Equ.(7);
8       end
9     end
10    Combine the learned representations from all
       meta-paths via Equ.(8);
11  end
12  Fuse the representations from all layers to generate  $\mathbf{h}_i^\Phi$  via
     Equ.(9);
13  for  $l \in L_\Psi$  do
14    Generate the node representation in the network
       schema channel via Equ.(11);
15  end
16  Fuse the representations from all layers to generate  $\mathbf{h}_i^\Psi$  via
     Equ.(12);
17 end
18 Extract the student's knowledge state via Equ.(15);
19 Capture the global intent representation  $\mathbf{H}_l$  via Equ.(22);
20 Capture the local intent representation  $\mathbf{H}_s$  via Equ.(26);
21 Fuse the global and local intents to generate  $\mathbf{V}_t$  via Equ.(28);
22 Predict the response  $r'_{t+1}$  via Equ.(29);
23 return  $r'_{t+1}$ .

```

4.1. Datasets

We evaluate different models on several typical knowledge tracing datasets, including ASSISTments2009,¹ ASSISTments2017,² EdNet,³ and JunyiAcademy⁴ which have been widely used in the educational data mining field for students' performance prediction.

ASSISTments2009 This dataset was collected from ASSISTments online education platform during the school year 2009–2010, which has two versions: Skill-builder and Non-Skill Builder. we select the updated Skill-builder version to conduct our experiments and remove duplicated records as the previous methods. Besides, we remove those students whose length of historical records is less than 3.

ASSISTments2017 This dataset was collected from the same tutoring platform as ASSISTments2009 and was used in Data Mining Competition 2017. It includes an 11-year data spanning from middle schools for mathematics. We remove the data recorded as 'noskill' in the skill column.

¹ <https://sites.google.com/site/assistmentsdata/home/assistent-2009-2010-data>.

² <https://sites.google.com/view/assistmentsdatamining/dataset>.

³ <https://github.com/riiid/ednet>.

⁴ <https://pslcdatashop.web.cmu.edu/Files?datasetId=1198>.

Table 1
Dataset statistics.

Dataset	Question	Attribute	Student	Record	Network schema	Meta-path
ASSISTments2009	17 751	Skill:123 Template:658	4151	325 637	Q-S: 21 246 Q-T: 17 751	Q-S-Q Q-T-Q
ASSISTments2017	2210	Skill:101 Assign:236	1709	942 816	Q-S: 2803 Q-A: 16 820	Q-S-Q Q-A-Q
EdNet	12 120	Skill:189 Bundle:8853	5000	6 212 117	Q-S: 28 758 Q-B: 13 169	Q-S-Q Q-B-Q
JunyiAcademy	509	Topic:39 Area:9	5000	346 918	Q-T: 509 Q-A: 509	Q-T-Q Q-A-Q

EdNet This dataset was collected from multi-platform AI tutoring records over 2 years with more than 780 k users in Korea, which contains various features of a student's actions, e.g., which learning material he has consumed, how much time he has spent on solving a given question or reading through experts' commentary. It should be noted that EdNet consists of over 130 million interactions collected from 784,309 students, making it too extensive for training purposes. Therefore, we randomly selected a sample of 5000 students and their corresponding interactive records to conduct our experiments.

JunyiAcademy This dataset includes problem logs and exercise-related information from Junyi Academy, along with exercise relationship annotations used for building models. Due to the size of the dataset, which includes over 25 million interactions from 247,606 students, we randomly selected a subset of 5000 students and their corresponding interactive records for our experiments.

The statistics of those datasets are listed in Table 1.

- Skill(S). The skill associated with the question. A question may cover multiple skills.
- Template(T). The template of the ASSISTments, ASSISTments with the same template ID have similar questions.
- Assign(A). A unique record is used for identifying an assignment with questions.
- Bundle (B). The expert-annotated bundle for each question, Each bundle can contain from 1 to 5 questions.
- Topic (T). The topic of each question represents a predominant concept in the knowledge map.
- Area (A). The knowledge area that question covers. Each area may contain several topics.

4.2. Baseline methods

To verify the effectiveness of our method, we implement some state-of-the-art methods as baselines for evaluation.

- *DKT* (Piech et al., 2015) uses the long short term memory (LSTM) to model the evolution of student state.
- *DKVMN* (Zhang et al., 2017) uses the dynamic memory network to store student's mastery of each skill.
- *GIKT* (Yang et al., 2020) generates question embeddings through fusing question-skill relations by GCN and applies a LSTM to update the student's latent knowledge state.
- *SAKT* (Pandey & Karypis, 2019) uses the attention mechanism to activate related historical interaction.
- *AKT* (Ghosh et al., 2020) uses the two-layer monotonic attention with rasch embeddings to model sequential interaction and activate related interaction.

4.3. Experimental settings

In our experiments, we set the input sequence length as 200. If a student has more than 200 records, we truncate the whole sequence into several sub-sequences. We choose the size of the hidden layer of RNN as $D_r = 320$ and the dilated coefficient as $\gamma = 2$. For the network schema channel of DHGN, we apply one-layer RGCN to aggregate the one-hop neighbors. As the variations of some parameters have great

impact on the results, we conduct the ablation study to pursue the optimal performance. The dimension of embeddings D is searched from $\{64, 128, 256, 320, 512\}$, the number of propagation layers in the meta-path channel L_ϕ from $\{1, 2, 4, 6, 8\}$ and the number of layers in local attention L_a from $\{1, 2, 3, 4, 6\}$ and the initial window size w_0 from $\{2, 4, 6, 8, 10\}$. The temperature parameter r is tuned within the range of $\{0.1, 0.2, 0.5, 1.0, 2.0\}$ and the hyperparameter λ is searched from the range of $\{0.01, 0.1, 0.5, 1, 10\}$. Each dataset is divided based on students, among which 80% of students are used for training and validating, and the rest 20% for testing. The Adam optimizer (Kingma & Ba, 2014) is applied to optimize the learnable parameters in the network with the learning rate and the batch size being tuned among $\{1e^{-3}, 5e^{-4}, 1e^{-4}\}$ and $\{64, 128, 256\}$, respectively.

4.4. Results

To follow the previous methods, we also evaluate the performance by the Area Under Curve (AUC). We compare our method with different baselines across all datasets. As shown in Table 2, HAN+S²-DHGN consistently outperforms other baselines and achieves state-of-the-art performance, which improves about 2.40%, 2.22%, 1.33%, 3.01% AUC scores over AKT and 4.31%, 3.72%, 2.42%, 2.69% over GIKT (60% training). When removing DHGN module (the initial question embedding h_i^0 is directly fed into the state extractor network), our model (HAN) still achieves performance comparable to AKT and improves about 4.36%, 9.97%, 8.70%, 7.03% over SAKT. When removing HAN module (the knowledge state x_i is directly fed into prediction network), our model (S²-DHGN) outperforms GIKT about 3.49%, 2.64%, 1.75%, 1.94%. Even without self-supervised learning, our proposed DHGN still performs best in most cases. The performance advantages confirm the superiority of the proposed DHGN in dealing with graph structure information. As the ratio of the training set decreases, the superiority of our model becomes more prominent, which also confirms the rationality and effectiveness of DHGN in handling the sparse data problem.

- Compared with traditional graph-based methods, the significant performance improvement of the proposed method indicates that fusing heterogeneous information is necessary to build representations for nodes, especially when the records are sparse. By capturing the high-order and local relations in the meta-path and network schema channels, respectively, our proposed method can learn richer semantics for the representations of questions.

Meanwhile, the self-supervised learning has great potential to further improve DHGN. Compared with the vanilla version, the self-supervised version shows decent improvements in all cases. When we reduce the percentage of training set to 20%, the improvement of S²-DHGN increases from 1.69%, 1.04%, 0.34%, 0.92% to 2.60%, 2.18%, 0.76%, 1.07% compared with DHGN, which is in line with our analysis that the sparsity of records might limit the learning of representation, and introducing the self-supervised task could alleviate the sparse data problem to some extent.

- Compared with traditional attention-based methods, the improvement of the proposed method demonstrates the necessity of integrating the long-term and short-term intents adaptively for student modeling. Through experiments, we find that the traditional attention mechanism,

Table 2

Performance (AUC) of various methods.

Dataset	ASSISTments 2009			ASSISTments 2017			Ednet			JunyiAcademy		
	20%	40%	60%	20%	40%	60%	20%	40%	60%	20%	40%	60%
DKT	0.7738	0.7920	0.8012	0.6904	0.7238	0.7473	0.7088	0.7229	0.7445	0.7105	0.7227	0.7443
DKVMN	0.7780	0.7924	0.8114	0.6753	0.7080	0.7406	0.6983	0.7198	0.7492	0.7083	0.7170	0.7349
GKT	0.7962	0.8089	0.8145	0.7189	0.7395	0.7581	0.7356	0.7490	0.7523	0.7300	0.7456	0.7611
SAKT	0.7402	0.7701	0.7932	0.6204	0.6531	0.7009	0.6534	0.6678	0.6989	0.6371	0.6833	0.7102
AKT	0.7830	0.8071	0.8297	0.7049	0.7438	0.7692	0.7164	0.7382	0.7604	0.7157	0.7402	0.7588
HAN	0.7846	0.8032	0.8278	0.7117	0.7481	0.7708	0.7203	0.7388	0.7597	0.7182	0.7416	0.7601
DHGN	0.8063	0.8205	0.8289	0.7291	0.7516	0.7701	0.7382	0.7533	0.7629	0.7487	0.7602	0.7688
S ² -DHGN	0.8273	0.8352	0.8429	0.7450	0.7562	0.7781	0.7438	0.7610	0.7655	0.7567	0.7653	0.7759
HAN + DHGN	0.8174	0.8277	0.8423	0.7382	0.7594	0.7805	0.7458	0.7629	0.7671	0.7563	0.7668	0.7770
HAN + S ² -DHGN	0.8321	0.8391	0.8496	0.7538	0.7689	0.7863	0.7490	0.7682	0.7705	0.7631	0.7708	0.7816

like SAKT, may not perform well in KT. Due to the long learning cycle, the forgetting behavior has a significant impact on a student's knowledge state. Attending relative interactions simply over long sequence is adverse to prediction. By equipping the attention network with decay coefficient and hierarchical local structure, our model can effectively track the student's intent in different terms and outperform those attention-based methods.

4.5. Ablation study

Here we conduct some ablation studies to demonstrate the contributions of each component of the proposed method. As the superiority of our model results from the joint effect of DHGN and HAN, we develop several variants for them and analyze their impact.

Analysis of DHGN. To investigate the effects of components of the proposed DHGN, we design several variants of DHGN with similar model complexity and design specifications as follows.

- DHGN-M: only uses the meta-path channel and aggregates heterogeneous neighbors' information by the proposed attention mechanism.
- GCN (Kipf & Welling, 2016)-M: adopts a graph convolutional network to aggregate neighbors' information in the meta-path channel. As it is designed for homogeneous graphs, we test all the meta-paths for GCN and show the optimal performance.
- GraphSAGE(mean/pooling) (Hamilton et al., 2017)-M: utilizes a modified mean/pooling-based aggregator to aggregate neighbors' information in the meta-path channel. We test both the mean variant and the pooling variant of GraphSAGE and report the optimal performance in the table.
- GAT (Veličković et al., 2017)-M: adopts a graph attention network to aggregate neighbors' information in the meta-path channel. Similarly, we test all the meta-paths for GAT and show the optimal performance.
- GIN (Xu et al., 2018)-M: applies a graph isomorphism network to aggregate neighbors' information in the meta-path channel. Note that the parameter ϵ of GIN is fixed at 0 in our experiments.
- HAN (Wang et al., 2019)-M: employs node-level and semantic-level attentions simultaneously on the heterogeneous graph, and assigns different importance to each meta-path.
- DHGN-S: only uses the network schema channel and aggregates neighbors' information by the relational graph convolutional network.
- GCN (Kipf & Welling, 2016)-S: adopts a graph convolutional network to aggregate neighbors' information in the network schema channel. We ignore the types of nodes and assign the same importance to different types of neighbors.
- GraphSAGE(mean/pooling) (Hamilton et al., 2017)-S: utilizes a modified mean/pooling-based aggregator to aggregate neighbors' information in the network schema channel. This configuration is the same as that of the meta-path channel.
- GAT (Veličković et al., 2017)-S: adopts a graph attention network to aggregate all types of neighbors' information in the network schema channel.

Table 3

Ablation study of DHGN.

Methods	ASSISTments2009	ASSISTments2017
DHGN-M	0.8266	0.7673
GCN-M	0.8175	0.7631
GraphSAGE(mean/pooling)-M	0.8181	0.7629
GAT-M	0.8207	0.7634
GIN-M	0.8152	0.7625
HAN-M	0.8228	0.7650
DHGN-S	0.8221	0.7640
GCN-S	0.8203	0.7623
GraphSAGE(mean/pooling)-S	0.8205	0.7622
GAT-S	0.8195	0.7615
GIN-S	0.8179	0.7607
HAN-S	0.8214	0.7628
DHGN	0.8289	0.7701

• GIN (Xu et al., 2018)-S: applies a graph isomorphism network to aggregate all types of neighbors' information in the network schema channel. The parameter ϵ of GIN is fixed at 0.

• HAN (Wang et al., 2019)-S: employs node-level and semantic-level attentions simultaneously to aggregate different types of neighbors' information in the network schema channel.

As shown in Table 3, DHGN outperforms other variants under similar model complexity and design specifications. Besides, the dual-channel structure improves model performance consistently on both datasets, which indicates the necessity of fusing the high-order and local relations to complement the representation of questions. When only using the meta-path channel, DHGN-M is still competitive, which demonstrates the effectiveness of modeling the heterogeneous correlations in the meta-path channel. Moreover, the decent gains also demonstrate the superiority of our proposed graph structure against those less powerful GNN variants in representation modeling.

Analysis of transformation functions. To investigate the positive contribution of α -entmax, we conduct experiments with the adaptive α and fixed α and compare their performance. The fixed α is chosen among {1, 1.2, 1.4, 1.6, 1.8, 2} and its value is the same for different meta-paths of DHGN. As shown in Table 4, the softmax function performs poor. A possible reason is that an excess of neighbors in the meta-path channel would lead to the semantic blending. The sparse transformation can alleviate the impact of unrelated nodes in the information aggregation and enforce the representation to focus on those highly correlated nodes. But the sparsimax does not always achieve optimal performance, which indicates that each meta-path may have its own best α . For different meta-paths, the concerned semantic relations are discrepant. We infer that the correlations between nodes might be affected by the intrinsic patterns in the meta-path. Just setting a fixed α for them disable it to reflect the discrepancy and learn informative representations for nodes. So we introduce the adaptive method to learn α based on the context in each meta-path. The superior performance of the adaptive method has further validated our analysis.

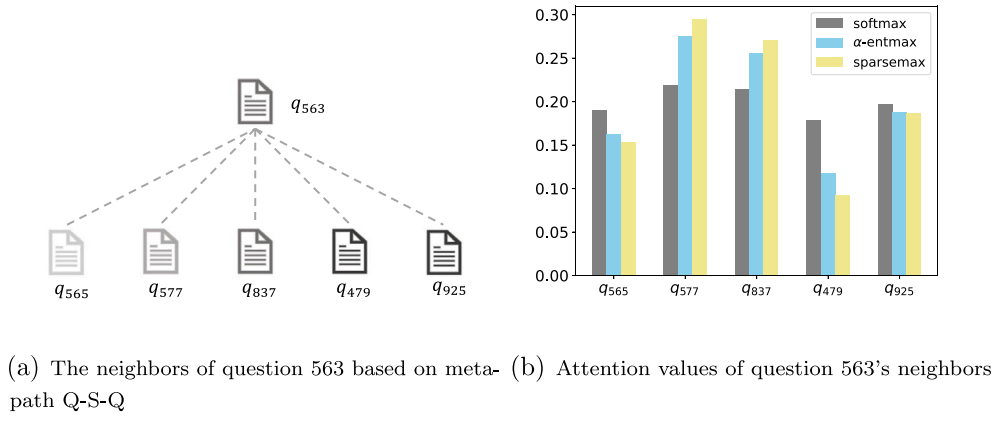


Fig. 5. Question 563's neighbors and corresponding attention values in DHGN. Softmax and sparsemax fix α at 1 and 2, respectively. α -entmax means applying the proposed adaptive α .

Table 4
Ablation study of transformation functions.

Methods	ASSISTments2009	ASSISTments2017
$\alpha = 1(\text{softmax})$	0.8205	0.7608
$\alpha = 1.2$	0.8198	0.7623
$\alpha = 1.4$	0.8211	0.7619
$\alpha = 1.6$	0.8247	0.7623
$\alpha = 1.8$	0.8238	0.7650
$\alpha = 2(\text{sparsemax})$	0.8234	0.7669
adap α	0.8289	0.7701

Table 5
Ablation study of HAN.

Methods	ASSISTments2009	ASSISTments2017
Local _s	0.8356	0.7758
Local _h	0.8401	0.7786
Global	0.8295	0.7683
Global _{η}	0.8302	0.7718
Global _{δ}	0.8370	0.7762
Global _{ζ}	0.8392	0.7781
Local _h + Global	0.8417	0.7805
Ours	0.8496	0.7863

Here we take question 563 of ASSISTments2009 dataset as an illustrative example. Given the meta-path Q-S-Q, which means that two different questions cover the same skill ("Probability of a Single Event"). We calculate the attention value of each neighbor under different transformation functions and show the results in Fig. 5(b). By the results, we can find that the proposed adaptive transformation tends to produce sparser weight distribution than softmax, which alleviates the semantic blending and helps improve representation quality to some extent.

Analysis of HAN. To verify the effects of each module of HAN, we consider several methods for comparison.

- Global, Local indicate only retaining global or local attention network in HAN, which demonstrates the different contributions of these two modules in state modeling.
- The subscripts of $\# \eta$, $\# \delta$ and $\# \zeta$ denote reserving the semantic-aware coefficient η , decay coefficient δ , refined coefficient ζ in the global attention network, respectively. No subscript indicates applying the standard attention without the above coefficients.
- The subscripts of $\# h$ and $\# s$ mean applying the proposed hierarchical and simple-layer structure in the local attention network, respectively. For fairness, we try the window size at each layer of hierarchical structure to conduct experiments for the simple-layer structure and select the optimal performance for comparison.

From Table 5, we observe that either removing global attention or local attention degrades the prediction performance, which proves

that they are complementary for each other. The hybrid structure, which fuses a student's long-term and short-term intents adaptively, is indispensable for modeling the knowledge state accurately. It is worth noting that the local attention shows better performance than the global attention in most cases. Based on this fact, we infer that modeling a student's knowledge state mainly relies on those questions that he recently interacted with. Focusing on those earlier interactions excessively may be adverse to prediction.

Moreover, the improvement of Local_h over Local_s confirms that the proposed hierarchical structure is more effective than simple-layer structure in perceiving a student's short-term intent. Besides, Global _{ζ} outperforms other variants, which also verifies the importance of retaining the low-level details and modeling the memory decay in the long-term intent.

In Fig. 6, we visualize the attention weights in the global attention and hierarchical local attention. Compared with the standard attention in Fig. 6(a), we can see that the proposed Global _{ζ} assigns lower weights to early interactions, which emphasizes the impact of memory decay on student's knowledge state and alleviates the negative effect of attending early interactions to some extent. Besides, we also can find that Local_h has smaller receptive fields than Global, which helps the model to focus on the recent interactive details and capture the student's short-term intent more precisely.

Analysis of hyper-parameters. To explore the stability of the proposed method, we further study the impact of some critical hyper-parameters and report the results in Fig. 7.

- Dimension of node embeddings. The results are shown in Fig. 7(a). When the dimension of h is set at 256, the model achieves the best performance. The performance trend of rising first and then dropping indicates that a smaller dimension may be inadequate to encode the semantic information while a larger dimension may cause additional redundancy for representation.

- Number of layers in the meta-path channel. The results are shown in Fig. 7(b). We can find that the performance grows with the number of layers when L_ϕ is less than 2. After that, the performance starts to drop due to over-smoothing.

- Number of layers in the local attention. We fix the size of the initial window w_0 at 4 and show the results with different numbers of layers in Fig. 7(c). A 3-layer setting achieves the best performance. As the number of layers is stacked (when L_a is larger than 3), the performance drops. A larger window does capture the relevance in a longer scope, but may introduce more unrelated interactions simultaneously and lead to suboptimal results.

- Size of the initial window in the local attention. Similar to the above, we fix the number of layers in the local attention L_a at 3 and show the results with different sizes of the initial window in Fig. 7(d). It can be observed that the best result is achieved when the size is 4. After

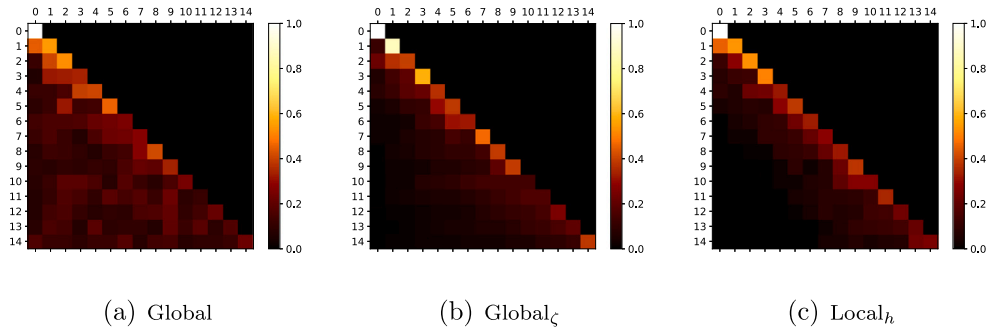


Fig. 6. Visualizations of attention weights in the global attention and the hierarchical local attention.

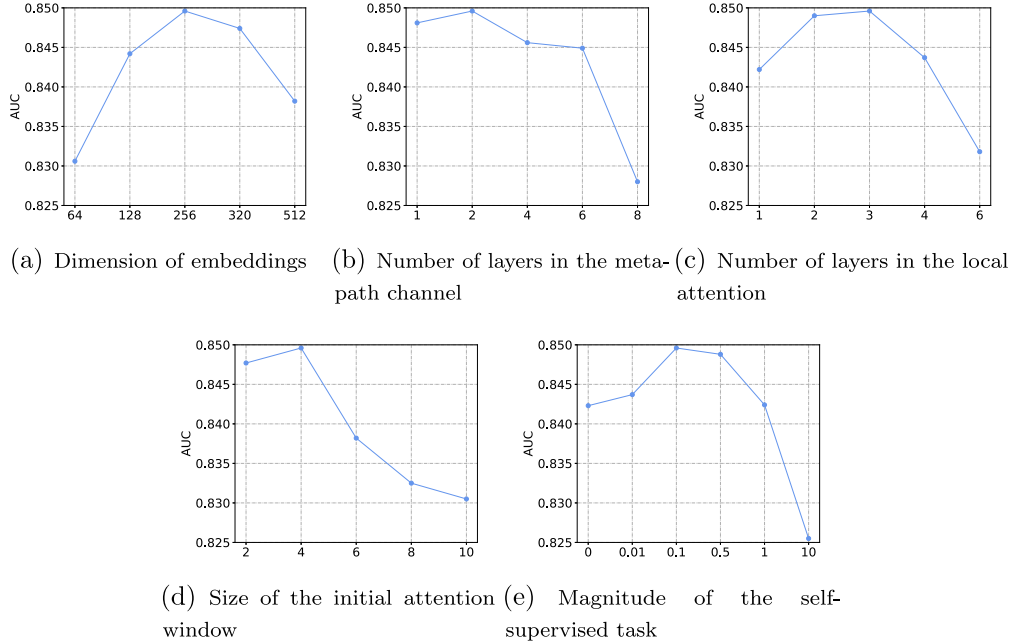


Fig. 7. AUC comparisons of each component with different parameters on ASSISTments2009.

that, the performance drops significantly, which indicates the local attention needs a suitable initial window to encode those interactions in the short term and a larger window may make it incapable of capturing the semantics details.

- Magnitude of the self-supervised task. The results are shown in Fig. 7(e). Note that the self-supervised task is removed when λ is set to 0. According to the results, we can find that the prediction task achieves decent gains with the self-supervised task when λ is less than 1. A larger λ would result in performance degradation. The possible reason is that a large weight would enforce the model to focus on auxiliary task excessively and the benefits brought by the self-supervised framework could be easily neutralized.

4.6. Visualization

For a more intuitive comparison, we apply t-SNE (Van der Maaten & Hinton, 2008) to visualize the question embeddings. We randomly select five pairs of ‘skill’ and ‘assign’ attributes, and those corresponding questions covering one pair of them in ASSISTments2017. The results are shown in Fig. 8, where each point represents one question. Note that, DKT, AKT, and GIKT (Fig. 8(a)–(c)), only consider the ‘skill’ as a single attribute of questions, and the color means the skill covered in this question. In our model, the color means one pair of attributes with this question. We can see that DKT does not perform well. Due to Rasch embeddings, AKT forms a curve for questions with the same

skills. However, for those multi-skill questions, AKT tends to form isolated spots, which is adverse to building representations. In GIKT, those questions covering different attributes are still mixed with each other to some extent. For the proposed DHGN, the distribution of nodes is more even and the boundaries of nodes with different attributes are relatively clearer, which demonstrates its effectiveness in keeping intra-class similarity.

5. Conclusion

In this paper, we propose a novel method, named fusing hybrid attentive network with self-supervised dual-channel heterogeneous graph, for student modeling and response prediction. We first propose a dual-channel heterogeneous graph network to capture the high-order and local relations in the meta-path and network schema channels, learning an informative representation for questions. To further enhance the representation, we supplement the network with a self-supervised task and contrast the views learned via the two channels. Then, in accordance with the historical interactive questions, we extract a student’s knowledge state by the parallel GRU. Last, in the proposed hybrid attention network, the student’s long-term and short-term intents, which are relative to the target question, are explicitly captured by the global and hierarchical local attention. Empirical results on several real-world datasets verify the effectiveness of our method.

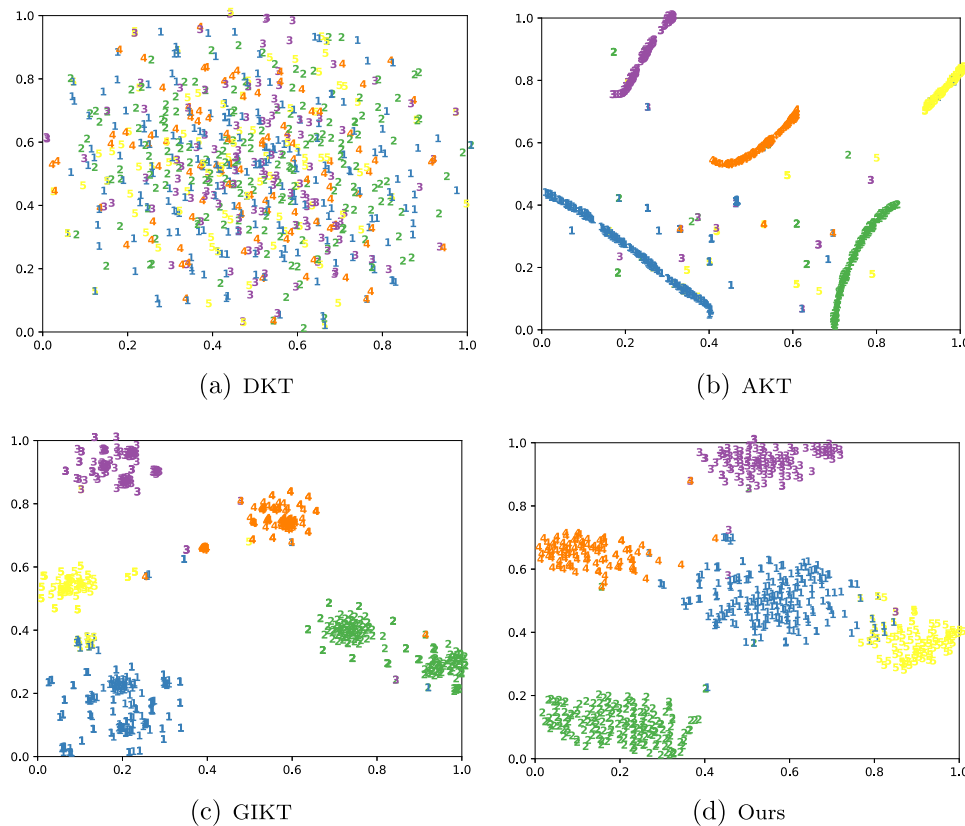


Fig. 8. Embedding visualization. Each point represents one question and its color stands for the attribute of the question.

However, there are still some limitations in our method. On one hand, the proposed graph network cannot effectively construct the representation of the newly updated questions in online systems. On the other hand, we capture a student's long-term and short-term intents in historical sequences by the proposed hybrid attention network, but ignore the differences of initial knowledge states of students. In the future, we plan to introduce a meta-learning paradigm to address the above limitations in the KT field, and exploit meta-learning at the KT model level and HIN at the data level simultaneously.

CRedit authorship contribution statement

Tangjie Wu: Conceptualization, Data curation, Software, Methodology, Formal analysis, Validation. **Qiang Ling:** Investigation, Project administration, Writing – review, Funding acquisition, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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